19. What happens to the domains in a ferromagnetic material in the presence of external magnetic field?
20. How is a galvanometer converted into (i) an ammeter and (ii) a voltmeter?

III Long answer questions

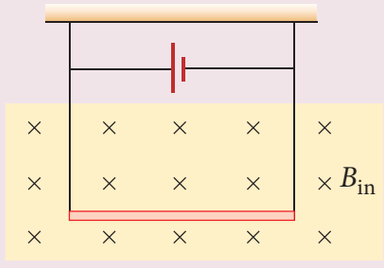
1. Discuss Earth's magnetic field in detail.
2. Deduce the relation for the magnetic field at a point due to an infinitely long straight conductor carrying current using Biot-Savart law.
3. Obtain a relation for the magnetic field at a point along the axis of a circular coil carrying current using Biot-Savart law.
4. Compute the torque experienced by a magnetic needle in a uniform magnetic field.
5. Calculate the magnetic field at a point on the axial line of a bar magnet.
6. Obtain the magnetic field at a point on the equatorial line of a bar magnet.
7. Find the magnetic field due to a long straight conductor using Ampere's circuital law.
8. Discuss the working of cyclotron in detail.
9. What is tangent law? Discuss in detail.
10. Derive the expression for the torque on a current-carrying coil in a magnetic field.
11. Discuss the conversion of galvanometer into an ammeter and also a voltmeter.
12. Calculate the magnetic field inside and outside of the long solenoid using Ampere's circuital law.
13. Derive the expression for the force between two parallel, current-carrying conductors.
14. Give an account of magnetic Lorentz force.
15. Compare the properties of soft and hard ferromagnetic materials.
16. Derive the expression for the force on a current-carrying conductor in a magnetic field.
17. Explain the principle and working of a moving coil galvanometer.

IV. Numerical problems

1. A bar magnet having a magnetic moment $\vec{p}_m$ is cut into four pieces i.e., first cut into two pieces along the axis of the magnet and each piece is further cut along the axis into two pieces. Compute the magnetic moment of each piece.

Answer $\vec{p}_{m_{new}} = \frac{1}{4}\vec{p}_m$

2. A conductor of linear mass density 0.2 g m^{-1} suspended by two flexible wire as shown in figure. Suppose the tension in the supporting wires is zero when it is kept inside the magnetic field of 1 T whose direction is into the page. Compute the current inside the conductor and also the direction of the current. Assume $g = 10$ m s^{-2}



Answer 2 mA

3. A circular coil with cross-sectional area 0.1 cm^2 is kept in a uniform magnetic field of strength 0.2 T. If the current passing in the coil is 3 A and plane of the loop is perpendicular to the direction of magnetic field. Calculate

12th Physics_Vol-1_Unit 3_English Medium.indd 192 23-12-2022 15:04:19